

Machine Learning Fundamentals

Custom knowledge-base material for the AI & Data Science RAG Chatbot project.

1. What is Machine Learning?

Machine learning allows systems to learn patterns from data and make predictions or decisions. Supervised learning uses labeled examples. Unsupervised learning works with data without target labels. Reinforcement learning learns through rewards and penalties.

2. Supervised Learning

Regression predicts continuous values, such as house prices. Classification predicts categories, such as spam or not spam. A feature is an input variable; the target or label is the value to predict.

3. Train, Validation, and Test

Training data is used to fit a model. Validation data helps choose settings and compare models. Test data is reserved for final evaluation. Data leakage occurs when information unavailable at prediction time enters training.

4. Linear and Logistic Regression

Linear regression models a continuous target as a weighted combination of features. Logistic regression estimates class probabilities, often using the sigmoid function for binary classification.

5. Tree Models and Ensembles

Decision trees split data using feature-based rules. Random Forest combines many trees trained with randomness to improve robustness. Gradient boosting builds models sequentially to correct earlier errors. XGBoost is a popular optimized gradient-boosting library.

6. Evaluation Metrics

For regression, common metrics are MAE, MSE, RMSE, and R-squared. For classification, accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix are common. The best metric depends on the problem and cost of errors.

7. Overfitting and Underfitting

Overfitting occurs when a model learns noise and performs poorly on new data. Underfitting occurs when the model is too simple. Regularization, cross-validation, more data, and appropriate model complexity can help.

8. Typical Workflow

Define the problem, collect data, clean and explore it, split data, preprocess features, train a baseline, evaluate, tune hyperparameters, inspect errors, and document limitations. Reproducibility requires fixed seeds and recorded configurations where appropriate.